

SOC Internship – Day 24 Notes

Ransomware Detection and Response

Objective

Learn how SOC analysts detect and respond to ransomware attacks.

1. What is Ransomware?

Ransomware is a type of malicious software (malware) that encrypts files or systems and demands payment (ransom) to restore access.

Main Goal of Attackers

- Encrypt important data
- Disrupt business operations
- Demand ransom payment

Common Targets

- Companies
- Hospitals
- Government organizations
- Educational institutions

2. Common Indicators of Ransomware

SOC analysts look for several warning signs:

- File encryption
- Suspicious PowerShell activity
- Disabled antivirus

- Mass file modifications

a) File Encryption

Large numbers of files suddenly become inaccessible or renamed.

Example Extensions

- .locked
- .encrypted
- .crypt

Risk

Loss of access to important files.

b) Suspicious PowerShell Activity

Attackers often use PowerShell scripts to execute ransomware silently.

Indicators

- Encoded commands
- Remote downloads
- Obfuscated scripts

Risk

Malware execution and lateral movement.

c) Disabled Antivirus

Ransomware may disable security tools before encrypting files.

Risk

System becomes unprotected against malware.

d) Mass File Modifications

Large numbers of files are modified within a short time.

Risk

Possible active encryption process.

3. Initial Access Methods

Attackers use different methods to enter systems:

- Phishing emails
- Weak passwords
- Exposed RDP
- Malicious downloads

a) Phishing Emails

Users receive malicious email attachments or links.

Example

Fake invoice or login page.

Risk

Malware infection.

b) Weak Passwords

Attackers guess or brute-force weak credentials.

Risk

Unauthorized system access.

c) Exposed RDP (Remote Desktop Protocol)

Internet-exposed RDP services are targeted by attackers.

Risk

Remote access to systems.

d) Malicious Downloads

Users unknowingly download infected software or files.

Risk

Direct malware execution.

4. Detection Techniques

SOC teams use several techniques to detect ransomware:

- Monitor abnormal file activity
- Detect encoded commands
- Identify backup deletion attempts

Abnormal File Activity Monitoring

Detect unusual file creation, renaming, or encryption behavior.

Encoded Command Detection

Identify suspicious PowerShell or command-line executions.

Example

```
powershell -EncodedCommand
```

Backup Deletion Detection

Attackers often delete backups before encryption.

Example Commands

```
vssadmin delete shadows
```

Risk

Prevents recovery of encrypted files.

5. Investigation Example

Alert

Large number of files encrypted.

Findings

- Encoded PowerShell execution
- Backup services disabled

Analysis

Possible ransomware attack detected.

6. Incident Response Steps

Step 1: Isolate Endpoint

Disconnect infected systems from the network.

Purpose

Prevent spread to other devices.

Step 2: Disable Affected Accounts

Disable compromised user accounts.

Purpose

Stop attacker access.

Step 3: Preserve Evidence

Collect logs, memory dumps, and forensic evidence.

Purpose

Support investigation and recovery.

Step 4: Begin Recovery

Restore systems using backups and recovery plans.

7. SOC Workflow

Endpoint Alert



SIEM Correlation



Detection



Containment



Recovery

Workflow Explanation

1. Endpoint security generates alert
2. SIEM correlates suspicious events
3. SOC confirms ransomware activity
4. Infected systems are contained
5. Recovery process begins

Importance of Backup Security

Best Practices

- Maintain offline backups
- Test backup recovery regularly
- Use strong authentication
- Restrict admin access
- Keep systems updated

Prevention Techniques

User Awareness Training

Educate users about phishing and malicious downloads.

Multi-Factor Authentication (MFA)

Add extra protection to accounts.

Patch Management

Keep systems and applications updated.

Network Segmentation

Limit ransomware spread across networks.

Assignment Answers

1. Explain Ransomware

Ransomware is malware that encrypts files or systems and demands payment from victims to restore access to their data.

2. List Indicators

Common Indicators

- File encryption
- Suspicious PowerShell activity
- Disabled antivirus
- Mass file modifications

- Backup deletion attempts
- Unusual network activity

3. Describe Response Steps

Incident Response Steps

1. Isolate infected endpoint
2. Disable affected accounts
3. Preserve forensic evidence
4. Contain the attack
5. Begin system recovery
6. Restore data from backups
7. Monitor for reinfection